

Anthony Bolivar

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EDUCATION

Colgate University

B.A. Computer Science (QuestBridge Scholar)

Aug. 2024 – Dec. 2027

Hamilton, NY

- **Relevant Coursework:** Data Structures & Algorithms, Introduction to Computer Systems, Discrete Structures, Linear Algebra

Yonsei University

Visiting Student, Computer Science (CIEE)

Aug. 2026 – Dec. 2026

Seoul, South Korea

- **Benjamin A. Gilman Scholarship (STEM)** — U.S. Dept. of State, 2026

EXPERIENCE

Software Engineering Intern

Octave (Hexagon AB)

May 2026 – Aug. 2026

Remote

- Planned and executed a migration of Octave's **Databridge Pro** from **Apache NiFi 2.4 to 3.0**, replacing NiFi Registry with a **Git/GitLab-based** version control system with **tenant isolation**, backed by **AWS S3, EC2, and Route 53**; shipped to production.
- Resolved **300+ Jira tickets** spanning bug fixes, emergency maintenance, and stack trace analysis across production incidents in a **multi-tenant, cloud-native Java** enterprise codebase on **AWS**.

Software Engineering Intern — Infrastructure & Automation

BMO Financial Group

Jun. 2025 – Aug. 2025

New York, NY

- Shipped Python and PowerShell automation services for health checks, log collection, and self-healing restarts across **3,000+ production endpoints**, reducing manual intervention time by **50%+**.
- Built idempotent, fault-tolerant provisioning pipelines supporting **1,500+ Active Directory accounts**, cutting setup time **80%** and configuration errors **90%**.
- Developed real-time monitoring dashboards over **10,000+ devices** with automated anomaly detection, surfacing failures and cutting mean troubleshooting time by **40%**.

Co-Founder & CTO

Raider Marketplace (Colgate TIA venture, LLC)

Jan. 2026 – Present

Hamilton, NY

- **Sole founding engineer.** Built the web app (**React/Vite**), iOS app (**Swift/SwiftUI**), and **Supabase/Postgres** backend with row-level security, Edge Functions, and a **Claude-powered AI moderation pipeline** gating every listing and seller profile. **Live on the App Store with 100+ active users.**
- Built and manage a small iOS engineering team; own code review, release coordination, and the shared-backend contract between web and mobile.
- Shipped real-time messaging (**Supabase Realtime**), personalized activity feed, branded business storefronts, service provider profiles, **APNs** push notifications, and automated Instagram card generation for approved listings.

PROJECTS

EU4-Haven | *Python, C++, PyTorch, Ghidra, WandB*

Feb. 2026 – Present

- Deep RL agent for *Europa Universalis IV* via runtime state extraction from a closed-source binary. **Behavioral cloning** on **6,600+ expert trajectories**, then **PPO** fine-tuned in **PyTorch** with a **14-component reward function** and dynamic masking over **145 discrete actions**; **247-dim observation space** fusing process memory, game state structs, and derived features. Trained on an NSF-funded HPC cluster.
- Built a **C++ DLL injection framework** (**IAT proxy hooking**, **VEH**, lock-free shared memory IPC) streaming live game state into Python training; reverse-engineered **80+ struct offsets** and **3,500+ annotated functions** via **Ghidra**.
- Designed an **LLM-assisted batch decompilation pipeline** (Claude API) reconstructing semantic layers of the closed-source Clausewitz engine from Ghidra disassembly — methodological contribution of an **arXiv preprint** in preparation.

TECHNICAL SKILLS

Languages: Python, C++, Java, TypeScript, Swift, C

ML & RL: PyTorch, NumPy, pandas, scikit-learn, WandB; PPO, behavioral cloning, dynamic action masking

Systems & Reverse Engineering: Ghidra, DLL injection (IAT, VEH), shared memory IPC, x86-64

Cloud & DevOps: AWS (EC2, S3, Route 53), Docker, Git, Maven, Jira, Vercel

Backend & Tools: FastAPI, Node.js, Next.js, PostgreSQL, Supabase, REST, SOAP, WebSocket, OAuth2

AI-Assisted Dev: Claude Code, Codex, Anthropic API

Spoken: English (native), Spanish (native), Japanese (intermediate), Korean (beginner)